18.4.2 Integrating constraints into the planning loop

ACME integrated this capability by adding a constraint-query interface to the existing authorization check.

During planning, the agent:

1. Identifies a potential action or resource.
2. Queries the policy system with a partial request.
3. Receives policy-derived constraints in the form of a policy residual.
4. Develops a plan within those constraints.

Later, during execution, each action still flows through the policy-aware loop described in section 18.3. The constraint query does not replace enforcement. It complements it.

This distinction is essential:

- Constraint queries guide planning.
- Authorization checks enforce decisions.

The policy engine is the authoritative source of truth in both cases.

18.4.3 Reducing trial and error

Consider an agent attempting to access a restricted directory containing credential files. In a purely reactive system, the agent may explore different ways to access the files—effectively probing for a path that works—only to discover through repeated denials that access is not permitted. Each denial forces the agent to revise its plan, resulting in repeated failures.

With constraint-aware planning, the agent queries the policy system first. A policy that restricts access to credential files returns a result indicating that such operations are not permitted. The agent incorporates this constraint into its reasoning and avoids attempting the operation entirely.